

NeuroDigital Interfaces and Adaptive Cognitive Networking: A Nanotechnology-Ready Framework for Human-AI Cognitive Interfaces

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ABSTRACT

NeuroDigital Interfaces (NDI) provide a systems pathway for translating non-invasive neural observations into machine-interpretable cognitive states while preserving uncertainty, privacy, and human agency. This paper presents NDAN, a NeuroDigital Adaptive Network that combines BCI signal preprocessing, probabilistic cognitive-state estimation, adaptive networking, and a governance layer for human-AI interaction. The proposed framework is nanotechnology-ready rather than nanotechnology-validated: nanosensors, flexible electrodes, and low-power edge devices are specified as future sensing substrates, while the current validation layer is software-based. We formalize the observation-to-intent pipeline, define uncertainty-aware state communication, and propose an evaluation protocol covering signal quality, latency, calibration, energy, privacy, and human-in-the-loop utility. The paper avoids claims of direct telepathy or clinical efficacy and instead establishes a reproducible architecture and translational roadmap for safe cognitive interfaces.

KEYWORDS

neurodigital interfaces, BCI, cognitive networking, nanosensors, human-AI interaction, neuro-rights

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1 INTRODUCTION

Human-AI interaction is increasingly constrained by the bandwidth and ambiguity of conventional input channels. NeuroDigital Interfaces can complement these channels by estimating task-relevant cognitive states from non-invasive measurements such as EEG or fNIRS. However, the engineering challenge is not simply signal

classification: a useful interface must quantify uncertainty, adapt to changing users and environments, communicate safely with distributed agents, and preserve autonomy.

This paper introduces NDAN as a system architecture rather than a claim of mind reading. The nanotechnology connection is defined as a future sensing and packaging layer involving nanosensors, flexible electrode interfaces, and low-power edge electronics. Those components require independent hardware and biomedical validation and are not presented as experimentally demonstrated here.

2 SYSTEM ARCHITECTURE

NDAN is organized into five layers:

- (1) **Sensing:** neural, physiological, and contextual observations;
- (2) **Signal processing:** filtering, artifact detection, feature extraction, and calibration;
- (3) **Cognitive state estimation:** probabilistic estimates of attention, workload, fatigue, and task intent;
- (4) **Adaptive networking:** state-aware communication among human, edge, and AI agents;
- (5) **Governance:** consent, provenance, access control, uncertainty disclosure, and reversibility.

The resulting loop is:

$$y_t \rightarrow \phi(y_t) \rightarrow p(z_t|\phi) \rightarrow m_t \rightarrow a_t \rightarrow y_{t+1}, \quad (1)$$

where y_t is a sensor observation, z_t is a latent cognitive state, m_t is a privacy-preserving message, and a_t is an adaptive system action.

3 NANOTECHNOLOGY-READY INTERFACE

The architecture identifies three interfaces where nanoscale engineering may provide future benefits:

- flexible, high-density electrode or nanosensor arrays;
- low-noise analog front ends and energy-aware edge processing;
- biocompatible packaging and reliable skin/interface contact.

These are requirements, not claims of a fabricated device. A valid hardware study must report material composition, sensor calibration, impedance, safety limits, sampling characteristics, and repeatability across users.

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4 COGNITIVE STATE COMMUNICATION

Rather than transmitting raw neural signals, NDAN transmits a state estimate and uncertainty:

$$m_t = (\hat{z}_t, \Sigma_t, c_t, \pi_t), \quad (2)$$

where $\hat{z}_t$ is the estimated state, Σ_t its uncertainty, c_t a consent scope, and π_t a provenance record. An agent may act only when confidence and consent satisfy policy constraints. This design reduces unnecessary data exposure and makes uncertainty a first-class networking object.

5 EVALUATION PROTOCOL

The proposed validation sequence contains four stages:

- (1) synthetic signals for unit and integration testing;
- (2) public EEG/BCI datasets with subject-wise splits;
- (3) controlled laboratory sessions with informed consent;
- (4) hardware-in-the-loop evaluation of the sensing and edge pipeline.

Metrics include balanced accuracy, calibration error, false activation rate, end-to-end latency, energy per inference, bandwidth per cognitive-state message, privacy leakage, and user-reported workload. No result should be generalized across users without subject-wise validation.

Table 1: Required reporting matrix for future validation.

Dimension	Required evidence
Signal	Sampling, filtering, artifacts, sensor calibration
Model	Architecture, training split, uncertainty method
Network	Message size, latency, loss, adaptation policy
Safety	Consent, withdrawal, privacy, failure-safe behavior
Hardware	Materials, impedance, power, thermal limits
Reproducibility	Seeds, code, dataset license, environment

6 GOVERNANCE AND NEURO-RIGHTS

NDAN must not infer sensitive attributes without explicit purpose and consent. The governance layer applies data minimization, local processing where possible, cryptographic provenance, user-visible uncertainty, and an emergency stop. A cognitive estimate is not a diagnosis, legal fact, or authorization to act. Human-in-the-loop review is mandatory for consequential actions.

7 DISCUSSION

NDAN connects nanoscale sensing, BCI signal processing, adaptive networks, and human-AI systems without collapsing them into an unsupported telepathy claim. The strongest near-term contribution is a reproducible systems architecture and evaluation framework. The most important research risk is domain shift: a model that works on synthetic signals may fail across users, electrode placements, motion artifacts, or changing tasks.

8 CONCLUSION

This paper defines a nanotechnology-ready architecture for NeuroDigital Interfaces and Adaptive Cognitive Networking. It provides a disciplined path from software prototypes to sensor and human-in-the-loop validation. Future work will implement public-dataset benchmarks, integrate the FCSTN/NDAN software lineage after its BCI test failures are corrected, and evaluate privacy-preserving cognitive-state communication.

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